

**FIFTH SEMESTER U.G. DEGREE (SPECIAL) EXAMINATION
NOVEMBER 2020**

(CUCBCSS—UG)

Mathematics

MAT 5D 18—MATHEMATICS FOR NATURAL SCIENCES

Time : Two Hours

Maximum : 40 Marks

Section A

*All questions to be attended.
Answer **all** the six questions.
Each question carries 1 mark.*

1. Write the relation between mean, median and mode.
2. Write $\log_b \left(\frac{xy^4}{z^3} \right)$ as the sum or difference of logarithms of x, y and z .
3. Find the harmonic mean H of the numbers 3, 5, 6, 6, 7, 10 and 12.
4. Define the semi-interquartile range.
5. Find the second and third and moments of the set 2, 3, 7, 8, 10.
6. Write the formula for finding skewness of a distribution.

(6 × 1 = 6 marks)

Section B

*All questions can be attended and overall ceiling.
Answer any **five** out of seven questions.
Each question carries 2 marks.*

7. Prove that the quadratic mean of two positive unequal numbers a and b is greater than their geometric mean.
8. If $Z_1 = X_1 + Y_1, Z_2 = X_2 + Y_2; \dots; Z_N = X_N + Y_N$: prove that $\bar{Z} = \bar{X} + \bar{Y}$.
9. Show that the product of the numbers 5.74 and 3.8, assumed to have three and two significant figures, respectively, cannot be accurate to more than two significant figures.

Turn over

10. The number of ATM transactions per day were recorded at 15 locations in a large city. The data were : 35, 49, 225, 50, 30, 65, 40, 55, 52, 76, 48, 325, 47, 32 and 60. Find : (a) the median number of transactions ; and (b) the mean number of transactions.
11. On a final examination in mathematics, the mean was 72 and the standard deviation was 15. Determine the standard scores (i.e., grades in standard-deviation units) of students receiving the grades : (a) 60 ; (b) 93 ; and (c) 72.
12. Prove that (a) $m_2 = m_2' - m_1'^2$; (b) $m_3 = m_3' - 3m_1' m_2' + 2m_1'^3$.
13. Prove that $w^2 + pw + q$, where p and q are given constants, is a minimum if and only if $w = \frac{-1}{2} p$.

(5 × 2 = 10 marks)

Section C

All questions can be attended and overall ceiling.

*Answer any **three** out of five questions.*

Each question carries 4 marks.

14. Solve the following logarithmic equation $\ln(5x) - \ln(4x + 2) = 4$.
15. The numbers $X_1, X_2, \dots, X_K$ occur with frequencies $f_1, f_2, \dots, f_K$ where $f_1 + f_2 + \dots + f_K = N$ is the total frequency.
- Find the geometric mean G of the numbers.
 - Derive an expression for $\log G$.
 - How can the results be used to find the geometric mean for data grouped into a frequency distribution?
16. Given the sets 2, 5, 8, 11, 14 and 2, 8, 14, find :
- The variance of each set.
 - The variance of the combined sets.
17. Find the (a) First ; (b) Second, (c) Third ; and (d) Fourth moments about the origin 4 for the set 2, 3, 7, 8, 10.
18. Find the : (a) Quartile ; and (b) Percentile coefficients of skewness for the distribution with $Q_1 = 268.25$, $Q_2 = 279.06$, $Q_3 = 290.75$.

(3 × 4 = 12 marks)

Section D

All questions can be attended and overall ceiling.

*Answer any **two** out of three questions.*

Each question carries 6 marks.

19. The final grades in Mathematics of 80 students at State University are recorded in the accompanying table :

68	84	75	82	68	90	62	88	76	93
73	79	88	73	60	93	71	59	85	75
61	65	75	87	74	62	95	78	63	72
66	78	82	75	94	77	69	74	68	60
96	78	89	61	75	95	60	79	83	71
79	62	67	97	78	85	76	65	71	75
65	80	73	57	88	78	62	76	53	74
86	67	73	81	72	63	76	75	85	77

With reference to this table, find :

- The highest grade.
- The lowest grade.
- The range.
- The grades of the five highest-ranking students.
- The grades of the five lowest-ranking students.
- The grade of the student ranking tenth highest.
- The number of students who received grades of 75 or higher.
- The number of students who received grades below 85.
- The percentage of students who received grades higher than 65 but not higher than 85.

Turn over

20. Find the mean deviation for the distribution of heights of 100 students in the following table :

<i>Height (in)</i>	<i>Frequency (f) :</i>
60 – 62	5
63 – 65	18
66 – 68	42
69 – 71	27
72 – 74	8

21. Calculate the first four moments about the mean for the distribution of the following table :

<i>X</i>	<i>f</i>
12	1
14	4
16	6
18	10
20	7
22	2

(2 × 6 = 12 marks)

FIFTH SEMESTER B.A./B.Sc. DEGREE EXAMINATION, NOVEMBER 2021

(CUCBCSS-UG)

Mathematics

MAT 5D 19—MATHEMATICS FOR SOCIAL SCIENCES

Time : Two Hours

Maximum : 40 Marks

Section A

*Answer all the six questions.
Each question carries 1 mark.*

1. Solve : $\frac{x}{6} - 5 = \frac{x}{9} + 1$.
2. Find $\lim_{x \rightarrow 4} \sqrt{2x^3 - 7}$.
3. Define concavity and convexity.
4. Find the partial derivative $\frac{\partial z}{\partial y}$ if $z = 8x^2 + 14xy + 5y^2$.
5. Find the marginal cost at $Q = 3$ if the total cost function $TC = 3Q^2 + 7Q + 12$.
6. Convert the logarithm $\log_a y = 5x$ into equivalent natural exponential form.

(6 × 1 = 6 marks)

Section B

*Answer any five out of seven questions.
Each question carries 2 marks.*

7. Verify whether the function $f(x) = \frac{x^2 + 3x + 12}{x - 3}$ is continuous at $x = 4$.
8. Use the point-slope formula to derive the equation of the line passing through (3, 11) and having slope - 4.
9. Find the derivative of $y = (3x^3 + 8)^5$.
10. Solve $7 \log_e x - 2.6 = 10$.
11. Integrate $\sqrt{3x^2 - 4}$ (6x) with respect to x .

Turn over

12. Evaluate $\int_1^3 (6x^2 + 5) dx$.
13. The marginal cost function of a firm is given by $MC = 3,000e^{0.3x} + 50$, when x is quantity produced. If fixed cost is Rs. 80,000, find the total cost function of the firm.
(5 × 2 = 10 marks)

Section C

Answer any **three** out of five questions.
Each question carries 4 marks.

14. If $y = x^{\log x}$, find $\frac{dy}{dx}$.
15. Evaluate $\lim_{x \rightarrow \infty} \sqrt{x} (\sqrt{x+4} - \sqrt{x})$.
16. Find the cross partial derivatives z_{xy} and z_{yx} for the function $z = 10(9x - 4y)^5$.
17. Find the relative extrema for the function $f(x) = -9x^2 + 126x - 45$.
18. The marginal revenue function of a product is given by $MR = 500 - 0.01x$ and the marginal cost function is given by $MC = 100 + 0.006x$. The fixed cost is Rs. 1,50,000. Find the profit function.
(3 × 4 = 12 marks)

Section D

Answer any **two** out of three questions.
Each question carries 6 marks.

19. Use integration by parts to find the indefinite integral $\int 30x\sqrt{9+x} dx$.
20. A firm has the demand function $P = 100 - 0.01Q$ and total cost function $TC = 50Q + 30,000$ where P is the price and Q is the number of units. How many units are to be manufactured to maximize the profit.
21. (a) Evaluate $\int \frac{xe^x}{(x+1)^2} dx$.
- (b) Find the equation of the line passing through the point (3, 1) and perpendicular to the line $2x + 7y - 5 = 0$.

(2 × 6 = 12 marks)

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(CUCBCSS—UG)

Mathematics

MAT 5D 18—MATHEMATICS FOR NATURAL SCIENCES

Time : Two Hours

Maximum : 40 Marks

Section A*Answer all the six questions.**Each question carries 1 mark.*

1. Define discrete data and continuous data with an example.
2. Find the log to the base 8 of 4096.
3. Define any two measures of central tendency.
4. Find the median of the set of numbers 5, 5, 7, 9, 11, 12, 15, and 18.
5. Find the quadratic mean of the numbers 3, 5, 6, 6, 7, 10, and 12.
6. Define skewness of a distribution.

(6 × 1 = 6 marks)

Section B*Answer any five out of seven questions.**Each question carries 2 marks.*

7. Solve the logarithmic equation $\ln(x)^2 - 1 = 0$.
8. Describe two graphic representations of frequency distributions.
9. Ten measurements of the diameter of a cylinder were recorded by a scientist as 3.88, 4.09, 3.92, 3.97, 4.02, 3.95, 4.03, 3.92, 3.98, and 4.06 centimeters (cm). Find the arithmetic mean of the measurements.
10. Find the mean and mode for the set of numbers 3, 5, 2, 6, 5, 9, 5, 2, 8, and 6.
11. Find the second and third moments of the set of numbers 2, 3, 7, 8, 10.
12. Find the standard deviation of the set of numbers 9, 3, 8, 8, 9, 8, 9, 18.
13. Prove that $m_2 = m'_2 - m_1'^2$.

(5 × 2 = 10 marks)

Turn over